TLS_Pie Raspberry Pi Setup Checklist

Use this checklist to set up the Raspberry Pi for the TLS_Pie lidar capture system.

1. Prepare the Pi

- Insert the microSD card with Raspberry Pi OS.
- Boot the Pi and log in.
- Connect the Pi to the network.
- Open a terminal.

1a. Optional: wireless setup with no monitor/keyboard/mouse

- In Raspberry Pi Imager settings (gear icon), before writing the SD card, enable SSH and enter your WiFi SSID/password.
- After boot, connect from your laptop: `ssh @.local` (PowerShell has SSH built in).
- Optional: enable VNC (`sudo raspi-config` -> Interface Options -> VNC) for a full desktop view.
- The Pi's WiFi (`wlan0`) is separate from the wired lidar link (`eth0`), so this does not interfere with capture.

2. Run the setup script

Run this command:

```
sudo bash /home/lipi/TLS-Pie/Raspberry\ Pie4/setup_tls_pie_pi.sh
```

If the project is not yet in /home/lipi/TLS-Pie, copy it there first.

3. Verify the required packages are installed

The script installs:

- python3-pip
- python3-dev
- python3-rpi.gpio
- git
- tcpdump
- openssh-server

4. Verify the project files are present

Make sure these files exist:

- /home/lipi/TLS-Pie/Raspberry Pie4/TLS-Pie/VLPrecord.sh
- /home/lipi/TLS-Pie/Raspberry Pie4/TLS-Pie/VLPbuttons.py
- /home/lipi/TLS-Pie/Raspberry Pie4/TLS-Pie/VLPwaitbutton.py

- /home/lipi/TLS-Pie/Raspberry Pie4/TLS-Pie/VLPstatussignal.py

5. Verify status return wiring

- Pi GPIO22 is connected through the level shifter to MicroView D4 (PISTATUS)
- Shared ground is connected between Pi and MicroView

6. Test the script manually

Run:

```
sudo /home/lipi/TLS-Pie/Raspberry\ Pie4/TLS-Pie/VLPrecord.sh eth0
```

Expected behavior:

- the script starts
- it waits for the Arduino trigger
- it creates a .pcap file in /home/lipi/velodyne
- MicroView shows READY before scan
- MicroView shows SCANNING during scan
- MicroView shows REC when Pi confirms recording

7. If the interface is not eth0

Check the interface name:

```
ip -br addr
```

Then run the script with the correct interface name.

8. Enable automatic startup

The setup script creates a startup launcher for the Pi desktop session.

Reboot the Pi:

```
sudo reboot
```

9. Confirm the system works

After reboot:

- verify the script starts automatically
- verify a .pcap file is created
- verify the Pi is waiting for the Arduino trigger
- verify no PI TIMEOUT message appears unless Pi recording ACK fails

10. Common problems

- Wrong network interface name
- Missing packages
- No common ground between Arduino and Pi
- No .pcap file created
- Missing GPIO22 -> D4 status return wire

11. Quick recovery commands

If needed, run:

```
sudo apt update
sudo apt install -y python3-pip python3-dev python3-rpi.gpio git tcpdump openssh-server
sudo pip3 install RPi.GPIO
```